DM Assignment 3　314553021 洪薇欣

GitHub : https://github.com/veldahung/DM2026-Assignment-3.git

Section 1: Preliminary Analysis

Data Structure Observations

Each training file contains a 5-minute wrist accelerometer recording summarized as 300 rows (one row per second), with six columns: mean_x, mean_y, mean_z, std_x, std_y, std_z. Each file belongs to a specific user folder (60 training users) and carries a single activity label for the entire recording. The task is therefore file-level multi-class classification (6 classes) rather than frame-level segmentation. The dataset comprises 11,020 training files and 6,849 test files.

Examining the raw data revealed several structural properties that directly informed method design:

Fixed-length sequences: Every file has exactly 300 rows (mean = 300.0, std = 0.0, min = 300.0, max = 300.0 as confirmed by the shape summary), meaning temporal modeling can treat each recording as a uniform-length time series of length 300. This makes both CNN-based sequence models and global statistical feature extractors straightforward to apply without variable-length padding complications.

Train files: 11020 Test files: 6849
Train shape per file:

	n_rows
count	11020.0
mean	300.0
std	0.0
min	300.0
25%	300.0
50%	300.0
75%	300.0
max	300.0

dtype: float64

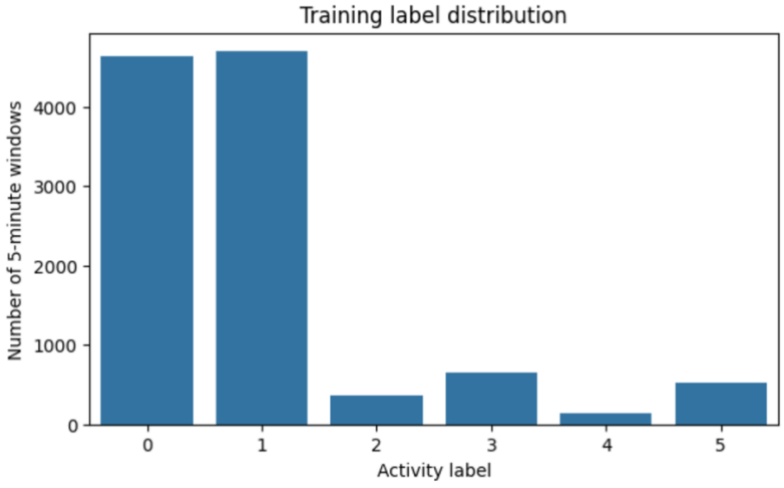
Test shape per file:

	n_rows
count	6849.0
mean	300.0
std	0.0
min	300.0
25%	300.0
50%	300.0
75%	300.0
max	300.0

Pre-summarized signal: The raw sensor readings are already aggregated into per-second mean and standard deviation. This means the input is not raw accelerometer samples at,

say, 50 Hz, but rather one-second summary statistics. Two important implications follow: (1) high-frequency FFT features computed on the 300-row sequence correspond to cycles over 300-second windows, not the original sensor frequency; (2) very short temporal patterns (sub-second motion bursts) are not recoverable from this representation.

Strong class imbalance: The training label distribution (see chart below) reveals severe imbalance across the 6 activity classes. Classes 0 and 1 each account for roughly 4,600+ samples and together dominate the training set, while classes 2, 3, 4, and 5 are all substantially underrepresented. The exact training counts are: class 0 — 4,643, class 1 — 4,695, class 2 — 358, class 3 — 656, class 4 — 142, class 5 — 526. Class 4 is the rarest with fewer than 150 samples — roughly 30× fewer than classes 0 and 1.



Class distribution:

label	count
0	4643
1	4695
2	358
3	656
4	142
5	526

This imbalance makes macro-F1 the appropriate metric rather than accuracy: a model that predicts only class 0 or 1 would achieve high accuracy but near-zero macro-F1. The minority classes (especially class 2 with 358 samples and class 4 with 142) become the dominant driver of score differences across methods, as even small absolute changes in their per-class F1 scores heavily move the macro average.

Waveform characteristics by class: Examining example 5-minute sequences for each label reveals qualitatively distinct temporal patterns across the six activity classes.

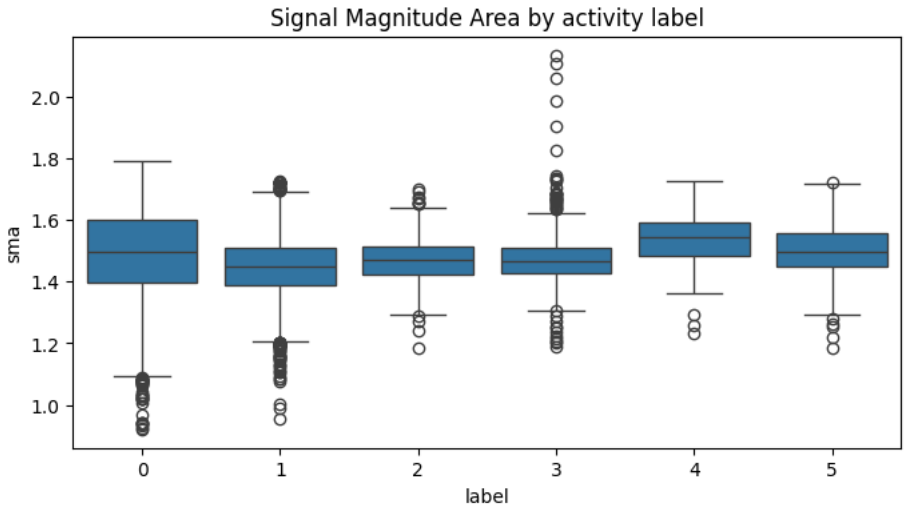
Label 0 shows near-flat, low-amplitude signals suggesting rest or a still posture. Labels 1 and 2 show moderate oscillation with occasional bursts. Label 3 exhibits high-amplitude, irregular multi-axis variation. Label 4 shows a mixed pattern with a transition partway through the recording. Label 5 has relatively stable mean values with a sharp step change visible around second 150 – 200, suggesting a discrete posture or orientation shift.

These waveforms confirm that the mean x/y/z channels encode both activity intensity and wrist orientation, and that temporal dynamics — including transitions, periodicity, and non-stationarity — vary substantially across classes.

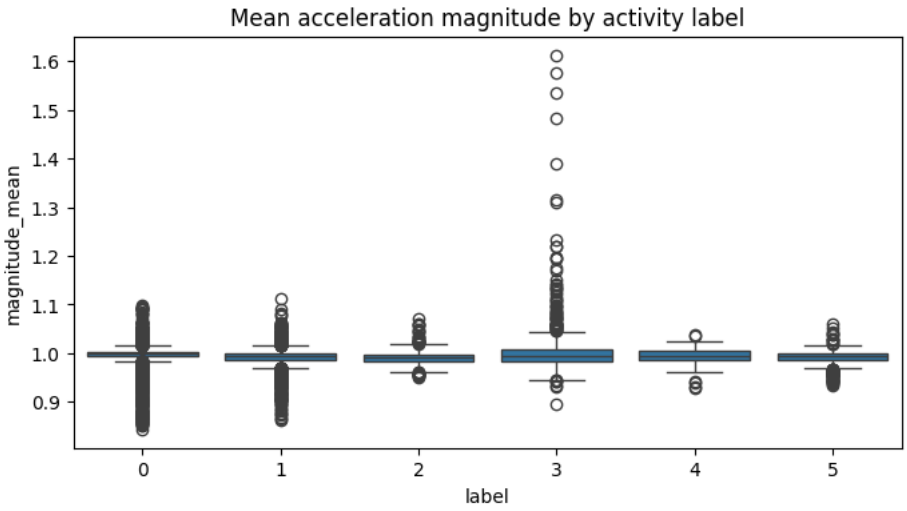
Signal Magnitude Area by class: The SMA (signal magnitude area, computed as the sum of absolute axis means per second) shows overlapping distributions across classes, with

class 0 having the widest spread (std = 0.150033) and class 2 the narrowest (std = 0.077149). Notably, class 3 has the most high-end outliers in SMA (values reaching above 2.0), consistent with the high-amplitude waveform observed in its time series.

The x-axis shows activity labels 0 – 5. All classes have median SMA near 1.4 – 1.5g. Class 0 has the largest interquartile range and many low outliers below 1.0. Class 3 has several extreme outliers above 1.8 – 2.2. Classes 2, 4, and 5 have tighter boxes. Overall, SMA alone does not cleanly separate classes, motivating richer feature engineering.



Mean acceleration magnitude by class: The mean acceleration magnitude (Euclidean norm of mean_x/y/z per file) is similarly uninformative on its own. All classes cluster near 1.0g (consistent with gravity being the dominant component in wrist accelerometry), with class 3 again showing the most outlier spread.



All six classes show median values tightly clustered around 1.0g. Class 3 has multiple extreme outliers above 1.3 – 1.6g. Classes 0 and 1 have tightly concentrated distributions with a few low-value outliers. Classes 2, 4, and 5 are compact. The plot illustrates that magnitude alone is not discriminative, reinforcing the need for temporal and distributional features.

The similarity of SMA and magnitude means across all six classes (all within ~0.05g of each other) confirms that the classification problem cannot be solved with simple intensity features alone. This directly motivated the development of richer temporal, frequency-domain, and distributional features in subsequent versions.

	sma		magnitude_mean	
	mean	std	mean	std
label				
0	1.485381	0.150033	0.995992	0.022489
1	1.447392	0.096836	0.991060	0.016821
2	1.470360	0.077149	0.992319	0.017354
3	1.474770	0.092160	1.005600	0.062055
4	1.537327	0.087320	0.993842	0.017460
5	1.503201	0.085656	0.991313	0.015139

Missing values: No missing values were found in either the training or test sets, so no imputation was required at any stage.

```
Missing values in train: 0
Missing values in test: 0
```

User-level variation: The 60 training users show substantial between-user variation in their sensor baselines. Wrist placement angle, wrist orientation, and individual motion style all shift the absolute values of mean_x/y/z. This was confirmed by the GroupKFold OOF drop: when validation folds contain entirely unseen users, OOF F1-macro fell from ~0.74 (StratifiedKFold) to ~0.71 (GroupKFold), a gap of ~0.03 attributable purely to user distribution shift.

Naive Baseline Performance

The RF-only submission (submission-2) serves as the naive baseline, measuring the ceiling achievable with global summary statistics alone and no temporal modeling, before any sequence model is introduced.

submission-2 (RF-only baseline, 0.7209): This was the first submission, labeled "test" on Kaggle. It was a pure Random Forest with no deep learning component, trained on 30 features per file: global mean, std, max, and min for each of the 6 axes, plus per-axis rolling-window means (window size 10 rows), and resultant magnitude mean/std. No cross-validation, no early stopping, no frequency-domain features. The rolling-window mean was the only temporal feature — it computed the average of a 10-second sliding window and then averaged that again over the full file, collapsing any temporal dynamics into a single scalar per axis.

submission-3 (RF + CNN-LSTM hybrid, 0.7537): The first hybrid ensemble, combining the same RF (on the same 26-column global-stat feature set, now without the rolling mean) with a CNN-LSTM deep learning model. The CNN-LSTM reshaped the 300-row sequence into 10 windows of 30 seconds each, processed each window with a TimeDistributed Conv1D(64, kernel=3) and GlobalAveragePooling, then passed the resulting 10-step sequence to an LSTM(64, dropout=0.2). Final predictions were a 50/50 probability average of RF and CNN-LSTM outputs. The model was trained for a fixed 20 epochs with batch size 16, no early stopping. The 0.7537 score was a substantial jump (+0.0328) over submission-2 purely from adding the CNN-LSTM temporal model.

Baseline deductions: The 0.7209 RF-only result confirms that global statistics alone are not sufficient: the macro-F1 ceiling for flat feature methods is limited by minority classes (especially 2 and 5), which require richer temporal and frequency-domain representations. The +0.0328 jump from adding CNN-LSTM over the RF alone demonstrates that the sequential 300-step structure contains discriminative temporal information that global statistics discard. This motivated systematic development of both richer hand-crafted features (FFT, autocorrelation, segment statistics) and improved deep learning architectures in subsequent versions.

Class Imbalance and Per-Class Difficulty

Two OOF classification reports were produced in the private experiments phase and together reveal the per-class difficulty across all method variants.

The GroupKFold LGB OOF report (macro-F1: 0.7123) and the user-median LGB OOF report (macro-F1: 0.7206) give slightly different per-class views due to the normalization applied in the latter.

```
OOF GroupKFold LightGBM macro-F1: 0.7123
Fold scores: [0.6521, 0.7409, 0.7251, 0.6843, 0.7024]
              precision    recall  f1-score   support

           0     0.9544    0.9698    0.9621      4643
           1     0.8817    0.9188    0.8999      4695
           2     0.3583    0.1201    0.1799       358
           3     0.6486    0.7287    0.6863       656
           4     0.8841    0.8592    0.8714       142
           5     0.7657    0.6027    0.6745       526

    accuracy                         0.8872     11020
   macro avg     0.7488    0.6999    0.7123     11020
weighted avg     0.8759    0.8872    0.8789     11020
```

```
Fold 5 user-median LGB macro-F1: 0.7077
OOF user-median LGB macro-F1: 0.7206
              precision    recall  f1-score   support

           0     0.9516    0.9696    0.9605      4643
           1     0.8831    0.9088    0.8958      4695
           2     0.3642    0.1648    0.2269       358
           3     0.6404    0.7683    0.6985       656
           4     0.8978    0.8662    0.8817       142
           5     0.7978    0.5627    0.6600       526

    accuracy                         0.8848     11020
   macro avg     0.7558    0.7068    0.7206     11020
weighted avg     0.8768    0.8848    0.8781     11020
```

The per-class difficulty pattern is consistent across both reports:

- Classes 0 and 1 (~4,643 and ~4,695 samples each): F1 scores of 0.9621 and 0.8999 (GroupKFold LGB) / 0.9605 and 0.8958 (user-median LGB). These are the easiest classes, likely corresponding to rest/still and a common everyday activity with high inter-user consistency.
- Class 4 (142 samples): Despite being the smallest class, it achieves F1 = 0.8714 (GroupKFold LGB) / 0.8817 (user-median LGB). Its distinctive sensor signature

(presumably a highly specific posture or vigorous activity) makes it relatively easy to identify even with few examples.

- Class 3 (656 samples): F1 = 0.6863 (GroupKFold LGB) / 0.6985 (user-median LGB). Confused with other classes frequently, though not catastrophically.
- Class 5 (526 samples): F1 = 0.6745 (GroupKFold LGB) / 0.6600 (user-median LGB), with recall of only 0.6027 / 0.5627 respectively. The model frequently misses true class-5 instances, suggesting high inter-user variability in how this activity manifests in 1-second summary statistics.
- Class 2 (358 samples): F1 = 0.1799 (GroupKFold LGB) / 0.2269 (user-median LGB), by far the worst. Precision 0.3583 / 0.3642 but recall only 0.1201 / 0.1648 — most true class-2 instances are classified as something else. No model variant solved this problem because the training signal is too weak and between-user variance too high for this class. The temperature scaling analysis confirmed this: class 2 had the highest temperature parameter (T = 1.131), meaning the model is overconfident on the few class-2 predictions it does make, yet still misses the vast majority.

Optimal temperatures per class:
[1.072, 1.086, 1.131, 1.014, 0.932, 1.078]
OOF macro-F1 before calibration: 0.7133 (RF+LGB tabular blend OOF)
OOF macro-F1 after calibration: 0.7129

Class 4 (T = 0.932) was the only under-confident class — the model was actually too conservative on its clearest signal. Temperature calibration had essentially no effect on label assignments (OOF F1 changed by only −0.0004), because the model's probability ordering was already well-ranked and the temperatures were all close to 1.0.

This per-class difficulty distribution means that virtually all macro-F1 score differences between submissions are driven by how well each method handles classes 2, 3, and 5 — not the majority classes.

Overall Submission Progression

The project evolved through three distinct phases, summarized below to provide context for answering question 2 – 4.

Phase 1 — Early exploration focused on establishing a working baseline and testing whether adding temporal or frequency-domain information to a simple Random Forest could improve activity recognition. The core question was whether the sequential structure of the 300-second recordings contained signal that flat global statistics discarded, and whether standard frequency features like FFT energy were a useful way to capture it.

Phase 2 — Architecture and ensemble upgrades shifted focus to replacing the shallow CNN-LSTM with a deeper residual sequence model, expanding the hand-crafted feature set substantially, and combining multiple tabular learners (Random Forest and LightGBM) with the CNN through soft-voting ensembles. This phase also introduced proper cross-validation and explored whether additional models (XGBoost, augmentation, more CNN seeds, higher CNN ensemble weight) could push performance further.

Phase 3 — Private generalization experiments / Generalization to unseen users was driven by the observation that strong public leaderboard scores did not reliably transfer to the private test set, where all users were completely unseen during training. Experiments focused on training and evaluating under GroupKFold by user identity, reducing the CNN's weight in the ensemble to limit user memorization, applying temperature calibration to the tabular models' probability outputs, and exploring user-level feature normalization to reduce sensor baseline drift across people.

Section 2: Preprocessing Techniques and Performance Impact

The following preprocessing techniques were applied across versions.

2.1 StandardScaler Normalization (all tabular versions)

All tabular features fed to Random Forest and LightGBM were standardized using `StandardScaler` (zero mean, unit variance) fitted on training data and applied identically to test data:

```
scaler = StandardScaler()
X_train_ml = scaler.fit_transform(train_feats)
X_test_ml  = scaler.transform(test_feats)
```

This is essential because the 69 – 133 engineered features span vastly different numeric scales: FFT spectral energy values can reach 1.94×10^7 (the top-variance feature across most rich-feature methods), while autocorrelation values lie in $[-1, 1]$ and tilt angles lie in $[-\pi/2, \pi/2]$. Without normalization, LightGBM's regularization terms `reg_alpha` and `reg_lambda` would apply unequally across features with very different scales, undermining its penalization of large weights. Random Forest is scale-invariant by nature — split scoring uses Gini impurity on value ordering, not absolute magnitude — so StandardScaler is applied uniformly to the full feature matrix for pipeline consistency rather than RF-specific benefit.

Impact: Applied in all versions from submission-2 onward. Its removal would degrade the LGB component substantially, but since it is present in every version it cannot be isolated as a standalone delta in the public score progression.

2.2 Per-Channel CNN Sequence Normalization (all CNN versions)

For the CNN, a per-feature-channel normalization was applied directly to the 300×6 sequence arrays after stacking:

```
mu    = X_train_seq.mean(axis=(0, 1), keepdims=True)
sigma = X_train_seq.std(axis=(0, 1),  keepdims=True) + 1e-7
X_train_seq = (X_train_seq - mu) / sigma
X_test_seq  = (X_test_seq  - mu) / sigma
```

Using axis=(0,1) computes one mean and one standard deviation per feature channel by pooling across all 11,020 training files and all 300 timesteps simultaneously, producing a keepdims shape of (1, 1, 6) that broadcasts correctly over the full array. This normalizes each of the 6 input channels (mean_x/y/z and std_x/y/z) independently to zero mean and unit variance. The 1e-7 term prevents division by zero on any constant channel. Per-channel normalization is critical because the mean channels center around gravity (~1 g) while the std channels are typically an order of magnitude smaller; a single global normalization across all channels would cause the network to effectively ignore the std

channels, discarding information about within-second motion variability that distinguishes classes like walking from standing.

Impact: Applied from submission-3 (first CNN version) onward. The CNN training behaviour table shows that the best Residual 1D-CNN (submission_improved) reached final train accuracy of 0.9187 and val accuracy of 0.9038 in run 1. Earlier CNN-LSTM versions used a single global mean/std across all channels and showed no val loss tracking, making a precise comparison imprecise, but the architectural improvements that accompanied this normalization change contributed to the +0.0233 gain from submission-3 to v3.

2.3 Constant-Length Zero-Padding (all CNN versions)

Before building the 300×6 sequence arrays, any file with fewer than 300 rows was zero-padded at the tail:

```
if len(sig) < 300:
    sig = np.pad(sig, ((0, 300 - len(sig)), (0, 0)), mode='constant')
```

Zero-padding at the tail preserves the actual signal at the head, which is appropriate since accelerometer activity is continuous throughout the recording window. Symmetric or edge-repeating padding would artificially inflate autocorrelation at lag 1 by duplicating the boundary values.

Impact: Virtually all files in this dataset are exactly 300 rows (confirmed by shape summary: mean=300.0, std=0.0). This guard rarely fires in practice but is required to avoid shape errors when batch-stacking sequences into the NumPy array that feeds the CNN.

2.4 Class-Balanced Training Weights (submission_improved onward)

Both the Random Forest and LightGBM models were trained with class_weight='balanced', and the CNN used the same mechanism via Keras's class_weight argument:

```
# Random Forest
```

```
rf = RandomForestClassifier(..., class_weight='balanced', ...)

# LightGBM
lgb_model = lgb.LGBMClassifier(..., class_weight='balanced', ...)

# CNN
cnn_model.fit(..., class_weight={i: w for i, w in enumerate(weights)})
```

The training set is severely imbalanced: classes 0 and 1 each have ~4,600 samples while class 4 has only 142 and class 2 has 358. Without reweighting, all three models would maximize accuracy by predicting majority classes, collapsing minority-class recall to near zero. Balanced weights rescale each sample's loss contribution inversely proportional to its class frequency, making a correct minority-class prediction roughly as influential on the loss as a batch of majority-class predictions.

Impact: The per-class F1 data shows a mixed but expected tradeoff. Class-2 F1 improved from 0.1266 (baseline RF, no weighting) to 0.2143 (best RF component with balanced weighting), and the LGB component reached 0.1798 with the 69-feature set and 0.2151 with the extended 133-feature set. However, class-5 F1 shifted from 0.6199 (baseline RF) to 0.5235 (best RF component). This reflects the inherent tension in balanced weighting: upweighting the rarest classes (2 and 4) redistributes model capacity away from moderately rare ones like class 5, which has more samples than class 2 but fewer than the majority classes. The net effect on macro-F1 is positive — the macro average weights all six classes equally, so large gains on the hardest minority classes (2, 4) outweigh the modest regression on class 5 — but balanced weighting is not a uniform improvement across all minority classes simultaneously.

2.5 EarlyStopping and ReduceLROnPlateau (submission_improved onward)

The Residual 1D-CNN introduced proper training callbacks replacing the earlier fixed-epoch CNN-LSTM:

```
callbacks = [
    tf.keras.callbacks.ReduceLROnPlateau(
        monitor='val_loss', patience=5, factor=0.5, min_lr=1e-5
    ),
    tf.keras.callbacks.EarlyStopping(
```

```
            monitor='val_loss', patience=12, restore_best_weights=True
        )
]
```

Earlier CNN-LSTM versions (submissions 3, 5, SMA) trained for a fixed 20 – 25 epochs with no validation split, so overfitting could not be detected or stopped. The CNN training behaviour table confirms the consequence: those models have no val_loss column and their overfitting status is listed as "unknown." The best Residual 1D-CNN (run 1) converged at epoch 34 (best epoch 22) with a mild train/val gap of 0.1081. Run 2 shows more overfitting (gap 0.2612), demonstrating that EarlyStopping does not eliminate seed-driven variance but does prevent runaway overfitting to the training set.

Impact: The CNN-LSTM → Residual 1D-CNN transition combined EarlyStopping with the architectural change. The combined switch from submission-3 to submission_v3 yielded +0.0233 on the public leaderboard (0.7537 → 0.7770). EarlyStopping accounts for some portion of this gain by selecting a better-generalizing weight checkpoint rather than the final epoch's weights.

2.6 Data Augmentation (submission_v2 only — negative result)

submission_v2 added two augmentation strategies to the CNN training pipeline: random time-scaling (stretching and compressing the 300-timestep sequence) and averaging two CNN seeds:

```
def augment(X, y, factor=2):
    """Jitter + time-shift augmentation to double training set."""
    ...
X_train_aug, y_train_aug = augment(X_train_seq, y_train_seq, factor=2)
```

The public score was 0.7676, below submission_v3's 0.7770 (no augmentation). On rerun, v2 dropped to 0.7526 ($\Delta = -0.0150$), while v3 only dropped to 0.7654 ($\Delta = -0.0116$). Augmentation increased variance rather than reducing it.

The reason is that the signal being augmented is not raw sensor samples but pre-aggregated one-second mean/std statistics. Stretching or compressing rows of summary statistics does not correspond to any realistic physical variation in wrist motion — a "compressed" recording of walking does not represent a person walking faster; it

represents a nonsensical interpolation of already-summarized statistics. The CNN trained on these synthetic examples learned spurious invariances that did not generalize to real test recordings.

Impact: Augmentation degraded public score by −0.0094 vs. v3 and increased rerun variance by 0.0034 (rerun drop of −0.0150 vs. −0.0116). It was not used in any subsequent submission.

2.7 Feature Engineering as the Primary Preprocessing Driver

The largest performance improvements across the project came from expanding the hand-crafted feature set. The feature matrix grew from 11,020×30 in the baseline to 11,020×69 in the best public submission and 11,020×133 in the extended private experiments.

Feature Group	Columns Added	Purpose
Global time-domain stats (mean, std, min, max)	24	Basic signal summary per axis
Extended stats: range, median, IQR, skew, kurtosis, energy, RMS, ZCR, p10/p90	30	Distribution shape and extremes
Autocorrelation at lags 1, 5, 10, 30	12	Periodic motion rhythm
Temporal segments (early/mid/late thirds)	18	Within-recording dynamics
FFT: peak frequency, spectral energy, entropy, top-3 components	20	Frequency-domain rhythm
Cross-axis: resultant magnitude stats, SMA	8	Orientation-invariant intensity
Pairwise axis correlations (xy, xz, yz), tilt angles (tilt_xy, tilt_yz)	5	Wrist orientation

The complete rich feature extraction function, implemented in extract_features(), combines all of the above groups for each of the three mean axes and three std axes. Key snippets illustrating each group:

```
# Autocorrelation at multiple lags
for lag in [1, 5, 10, 30]:
```

```
        feats[f'm_{ax}_autocorr_lag{lag}'] = float(pd.Series(m).autocorr(lag=lag))

# Temporal thirds
thirds = np.array_split(m, 3)
for t_idx, seg in enumerate(thirds):
        feats[f'm_{ax}_seg{t_idx}_mean'] = seg.mean()
        feats[f'm_{ax}_seg{t_idx}_std']  = seg.std()

# FFT dominant frequencies
fft_vals = np.abs(fft(m))[1:len(m) // 2]
feats[f'fft_{ax}_peak_freq']        = int(np.argmax(fft_vals)) + 1
feats[f'fft_{ax}_spectral_energy']  = float((fft_vals ** 2).sum())
feats[f'fft_{ax}_spectral_entropy'] = float(stats.entropy(fft_safe / fft_safe.sum()))

# Tilt angles
feats['tilt_xy'] = float(np.arctan2(mean_x_g, np.sqrt(mean_y_g**2 + mean_z_g**2)))

# Pairwise axis correlations
feats['corr_xz'] = float(df['mean_x'].corr(df['mean_z']))
```

The top variance features across rich-feature methods confirm that FFT spectral energy dominates the feature space (fft_y_spectral_energy: variance 1.94×10^7, fft_z_spectral_energy: 1.86×10^7, fft_x_spectral_energy: 1.45×10^7). The most important features by model agree: s_y_mean is the top RF importance (0.085) and top LGB importance (5,138 gain) across the best submission, followed by s_x_mean and s_z_mean — the per-second standard deviation of the mean channels, which captures motion variability, not just average level.

Feature set statistics also confirm data quality. Across all methods, the feature matrices contained zero NaN or Inf values after the .fillna(0) guard applied to any autocorrelation result that returned NaN on short or constant signals.

Performance impact by feature group addition:

Configuration	Public Score	Delta
Baseline RF, 30 features (global stats + rolling mean)	0.7209	—
RF + CNN-LSTM, same global-stat features	0.7537	+0.0328
RF + CNN-LSTM + FFT energy/max/mean	0.7227	−0.0310

RF + CNN-LSTM + SMA + peak frequency + spectral entropy	0.7475	+0.0248
RF + LGB + Residual 1D-CNN, full rich feature set	0.7881	+0.0406

Adding raw FFT energy/max/mean to the CNN-LSTM baseline degraded performance by −0.0310. These coarse spectral summaries are dominated by low-frequency components, are sensitive to between-user baseline drift, and are partially redundant with what the CNN already captures from the raw sequence — adding them provided no independent signal while diluting the RF's existing decision boundaries.

Switching to SMA + peak frequency + spectral entropy recovered +0.0248 because peak frequency (dominant repetition rate) and spectral entropy (spectrum concentration vs. diffuseness) are more user-invariant than total spectral energy: they identify the rhythm of an activity regardless of the user's wrist placement angle.

The largest single gain (+0.0406) came from the complete pipeline rebuild in submission_improved, where the full rich feature set was combined with LightGBM and a Residual 1D-CNN. These three changes cannot be individually isolated from the public score table, but the feature importance data confirms that the new features (s_y/x/z_mean, corr_xz, m_y_min) became the dominant predictors — features that were absent from the baseline's 30-column set.

2.8 User-Median Normalization (private experiments — negative result)

In Phase 3 private experiments, user-level median normalization was added to reduce inter-user sensor baseline drift caused by different wrist-wearing angles:

```
baseline = user_median_dict[col]      # median of all training files for this user
feats[f'{col}_median_norm'] = v - baseline
```

The normalized feature set (11,020×112, "Person-independent LightGBM normalized features") produced a public score of 0.7557 — lower than the GroupKFold LGB-only baseline of 0.7717. The failure stems from a train/test identity mismatch: during training, each file is normalized by its known user's median; at test time, user identities are unknown, so a global median across all 60 training users is used instead. This creates a systematic feature distribution shift at inference that does not exist during training. The per-fold OOF variation was high (0.6415 vs. 0.7520 across folds), confirming that some

user groups are harder to generalize across and that median normalization alone cannot close that gap without knowing test user identities.

2.9 Temperature Scaling (private experiments — negligible effect)

Per-class temperature scaling was applied to calibrate the tabular ensemble's probability outputs by minimizing negative log-likelihood on OOF predictions:

```
scaled_test_logits = log_test / T_opt      # T_opt per class, learned from OOF
```

Optimal temperatures were [1.072, 1.086, 1.131, 1.014, 0.932, 1.078]. All classes except class 4 (T = 0.932) were mildly overconfident (T > 1 softens probabilities). Class 2 had the highest temperature (T = 1.131), confirming that on the rare occasions the model predicted class 2, it was overconfident, yet still missed most true class-2 instances.

Impact: The RF+LGB tabular blend OOF macro-F1 changed from 0.7133 to 0.7129 after calibration — a negligible −0.0004. The public score was 0.7655. Since all temperatures were close to 1.0, rescaling did not change the argmax of any prediction. Calibration would only matter if the model's probability ordering were already wrong, which it was not here.

Section 3: Aligning Activity Labels with Sequential Accelerometer Readings

The Alignment Problem

Each file contains a 5-minute wrist accelerometer recording summarized as 300 rows (one row per second), and carries a single activity label for the entire recording. There is no frame-level annotation. The classification task is therefore to aggregate the sequential temporal information in 300 rows into a single prediction of which of 6 activity classes the file belongs to. Two parallel approaches were used to solve this alignment problem, combined via soft-voting ensemble.

Approach A: Global Statistical Features (RF and LightGBM)

The tabular models do not process the sequence directly. Instead, temporal dependencies are captured by summarizing the 300-row sequence into a fixed-length feature vector, encoding rhythm, dynamics, and within-recording change as scalars.

Autocorrelation at multiple lags: For each mean axis and lags [1, 5, 10, 30]:

```
feats[f'm_{ax}_autocorr_lag{lag}'] = float(pd.Series(m).autocorr(lag=lag))
```

Autocorrelation at lag 1 captures second-to-second smoothness: high for slow continuous activities (standing, sitting), low for rapid activities with irregular motion bursts. Lag 10 captures medium-term periodicity — a walking step cycle occurs at roughly 1 – 2 seconds, so lags 1 – 2 should be high for walking; stair climbing has a different rhythm. Lag 30 captures coarser repetition. Together these four lags give each tabular model a rhythm fingerprint per axis, encoding the temporal autocorrelation structure without requiring the model to process the raw sequence.

FFT dominant frequencies: For each mean axis:

```
fft_vals = np.abs(fft(m))[1:len(m) // 2]
feats[f'fft_{ax}_peak_freq']        = int(np.argmax(fft_vals)) + 1
feats[f'fft_{ax}_spectral_energy']  = float((fft_vals ** 2).sum())
feats[f'fft_{ax}_spectral_entropy'] = float(stats.entropy(fft_safe / fft_safe.sum()))

# Top-3 dominant frequencies
top3_idx = np.argsort(fft_vals)[-3:]
for rank, idx in enumerate(sorted(top3_idx)):
    feats[f'fft_{ax}_top{rank+1}_freq'] = int(idx) + 1
    feats[f'fft_{ax}_top{rank+1}_amp']  = float(fft_vals[idx])
```

The dominant frequency directly encodes the repetition rate of periodic activity. Because the sequence is sampled at 1 Hz with 300 points, the frequency axis spans 0 – 0.5 Hz; a walking activity at ~1.5 steps/second would alias to the low-frequency bins available in this 1 Hz sampling. Spectral entropy captures whether the spectrum is concentrated at one frequency (periodic activity like walking) or diffuse across all bins (irregular motion or

rest). Top-3 components capture multi-harmonic activities that produce energy at several related frequencies simultaneously.

Temporal segment statistics.: The 300-row sequence is divided into three equal segments (seconds 0 – 99, 100 – 199, 200 – 299):

```
python
thirds = np.array_split(m, 3)
for t_idx, seg in enumerate(thirds):
        feats[f'm_{ax}_seg{t_idx}_mean'] = seg.mean()
        feats[f'm_{ax}_seg{t_idx}_std']   = seg.std()
```

This captures non-stationarity within the recording. If an activity changes partway through the 5-minute window — as in class 4, which shows a visible transition in the raw waveform, and class 5, which has a sharp step change around second 150 – 200 — the difference between early and late segment means will be large. Activities with consistent behavior produce similar statistics across all three segments. The segment difference is a tabular proxy for temporal change without requiring sequence modeling.

Cross-axis correlations and tilt angles.: Pairwise Pearson correlations between axes encode motion geometry:

```
feats['corr_xy'] = float(df['mean_x'].corr(df['mean_y']))
feats['corr_xz'] = float(df['mean_x'].corr(df['mean_z']))
feats['corr_yz'] = float(df['mean_y'].corr(df['mean_z']))

feats['tilt_xy'] = float(np.arctan2(mean_x_g, np.sqrt(mean_y_g**2 + mean_z_g**2)))
feats['tilt_yz'] = float(np.arctan2(mean_y_g, np.sqrt(mean_x_g**2 + mean_z_g**2)))
```

Walking couples vertical and horizontal axes in a specific phase relationship, while stair climbing produces a different coupling due to the elevation component. corr_xz appears in the top-5 LGB feature importances for the best submission (3,029 gain), confirming it carries discriminative signal beyond what per-axis statistics alone provide. Tilt angles capture the average wrist orientation during the recording: static activities (sitting, standing) produce stable tilt angles consistent with the user's resting wrist posture, while dynamic activities produce variable tilt that averages to a different value.

Feature importance evidence for temporal features: The LGB feature importances for the best submission show that the top five features are s_y_mean (5,138), s_x_mean (3,772), s_z_mean (3,204), corr_xz (3,029), and m_y_min (2,973). The s_ prefix denotes per-

second standard deviation statistics — the mean of the std_axis columns over the 300 seconds. This feature directly encodes average motion variability per second, which is a summary of the temporal dynamics of within-second motion intensity. The cross-axis correlation and per-axis minimum round out the top five, confirming that temporal and cross-axis features drive the LGB component's predictions more than simple global means.

Approach B: Direct Sequence Modeling (CNN Architectures)

The CNN processes the full 300×6 sequence directly, learning temporal representations end-to-end without requiring handcrafted features.

CNN-LSTM (submissions 3, 5, SMA). The 300-row sequence was reshaped into 10 windows of 30 timesteps each. A TimeDistributed Conv1D(64, kernel=3) + BatchNorm + GlobalAveragePooling extracted local features within each 30-second window, and an LSTM(64, dropout=0.2) captured dependencies across the 10 windows:

```
inp = layers.Input(shape=(SEQ_LEN, WINDOW_SEC, len(FEATURE_COLS)))
x = layers.TimeDistributed(layers.Conv1D(64, 3, activation='relu', padding='same'))(inp)
x = layers.TimeDistributed(layers.BatchNormalization())(x)
x = layers.TimeDistributed(layers.GlobalAveragePooling1D())(x)
x = layers.LSTM(64, dropout=0.2)(x)
out = layers.Dense(n_classes, activation='softmax')(x)
```

This architecture captures temporal progression at a coarse scale only: the LSTM sees 10 context points corresponding to 30-second segments. Any pattern shorter than ~30 seconds is averaged away by GlobalAveragePooling before the LSTM sees it. The early CNN-LSTM versions also had no validation split and trained for a fixed 20 – 25 epochs, so no early stopping was possible.

Residual 1D-CNN (submission_improved onward). The flat 300×6 sequence was processed by a residual CNN with a Conv1D(64, kernel=7) stem followed by three residual blocks with filters [64, 128, 128], each using two Conv1D(5) layers with BatchNorm, residual skip connections, MaxPooling(2), and Dropout(0.2):

```
inp = layers.Input(shape=(seq_len, n_feats))
x = layers.Conv1D(64, 7, padding='same', activation='relu')(inp)
x = layers.BatchNormalization()(x)
```

```
for filters in [64, 128, 128]:
    res = layers.Conv1D(filters, 1, padding='same')(x)          # skip connection
    x   = layers.Conv1D(filters, 5, padding='same', activation='relu')(x)
    x   = layers.BatchNormalization()(x)
    x   = layers.Conv1D(filters, 5, padding='same', activation='relu')(x)
    x   = layers.BatchNormalization()(x)
    x   = layers.Add()([x, res])                                # residual
merge
    x   = layers.MaxPooling1D(2)(x)
    x   = layers.Dropout(0.2)(x)

x   = layers.GlobalAveragePooling1D()(x)
```

This is the key alignment mechanism for the CNN. GlobalAveragePooling1D aggregates the learned temporal feature map across all remaining timesteps (after three rounds of MaxPooling the 300-step sequence is reduced to ~37 steps) into a single fixed-length vector. This vector is the CNN's learned temporal summary of the entire recording, which then feeds the classification head. Averaging over all timesteps means the prediction reflects the typical temporal pattern throughout the 5 minutes, not just a single moment.

The private-experiment version expanded filters to [128, 256, 256] and added multi-scale global pooling:

```
avg_pool = layers.GlobalAveragePooling1D()(x)   # average activity level
max_pool = layers.GlobalMaxPooling1D()(x)       # peak activity moments
x = layers.Concatenate()([avg_pool, max_pool])
```

Concatenating average and max pooling allows the classification head to use both the typical temporal pattern and the most distinctive moment in the recording. The max pool is particularly useful for activities with brief but characteristic motion bursts that would be diluted by averaging alone (e.g., class 4's transition event partway through the recording).

Residual skip connections allow the network to combine low-level (raw motion amplitude) and high-level (multi-second pattern) features at all depths, and prevent vanishing gradients on the 300-step input that would limit plain deep CNNs. The kernel-7 stem captures patterns spanning 7 seconds before the residual blocks refine them to finer structure.

CNN Training Behaviour (R1 = first try; R2 = re-run)

Method	Run	Final train loss	Final val loss	Train/val gap	Final train acc	Final val acc	Epochs to converge	Overfit?
Baseline RF	—	—	—	—	—	—	—	—
Hybrid RF + CNN-LSTM	R1	0.3394	— (no val)	—	0.8789	—	20/20	unknown
Hybrid RF + CNN-LSTM	R2	0.3432	— (no val)	—	0.8825	—	20/20	unknown
FFT + CNN-LSTM	R1	0.3374	— (no val)	—	0.8835	—	20/20	unknown
FFT + CNN-LSTM	R2	0.3414	— (no val)	—	0.8820	—	20/20	unknown
SMA + peak-frequency + CNN-LSTM	R1	0.3495	— (no val)	—	0.8799	—	25/25	unknown
SMA + peak-frequency + CNN-LSTM	R2	0.3529	— (no val)	—	0.8776	—	25/25	unknown
BEST RF + LGB + Residual CNN	R1	0.2256	0.3337	0.1081	0.9187	0.9038	34 (early stop, best ep.22)	Mild
BEST RF + LGB +	R2	0.1425	0.4037	0.2612	0.9465	0.8848	31 (early	Yes

Residual CNN							stop, best ep.19)	
LGB + Residual CNN + augmented CNN (run 1)	R1	0.0057	0.0900	0.0843	0.9980	0.9746	60/60 (best ep.57)	Yes
LGB + Residual CNN + augmented CNN (run 2)	R1	0.2069	0.2441	0.0372	0.9260	0.9265	12 (early stop, best ep.1)	Minimal
LGB + Residual CNN + augmented CNN (run 1)	R2	0.0060	0.1248	0.1188	0.9979	0.9710	53 (early stop, best ep.41)	Yes
LGB + Residual CNN + augmented CNN (run 2)	R2	0.2101	0.2454	0.0353	0.9251	0.9251	12 (early stop, best ep.1)	Mild
LightGBM 5-fold + single CNN	R1	0.1220	0.4247	0.3027	0.9541	0.8929	34 (early stop, best ep.22)	Yes
LightGBM 5-fold + single CNN	R2	0.1195	0.4920	0.3725	0.9589	0.8757	26 (early stop, best ep.14)	Yes

RF + LGB + Residual CNN weight boosted	R1	0.1474	0.4928	0.3454	0.9437	0.8956	26 (early stop, best ep.14)	Yes
RF + LGB + Residual CNN weight boosted	R2	0.1443	0.5092	0.3649	0.9459	0.8866	24 (early stop, best ep.12)	Yes
CNN seed search	R1	best seed=2: val_loss=0.3551, val_acc=0.8966						Mild
CNN seed search	R2	best seed=0: val_loss=0.3425, val_acc=0.8993					11 epochs shown	Mild
Person-independent LightGBM normalized features	R1/R2	0.2613 (ep.11)	0.4373	0.1760	0.9093	0.8893	11 (partial)	Mild
Bigger CNN + label smoothing + spatial dropout	R1	best seed=42: val_loss=0.3516, val_acc=0.9065						Mild
Bigger CNN +	R2	best seed=0: val_loss=0.28						Mild

label smoothing + spatial dropout		98, val_acc=0.9138						

Approach C: Soft-Voting Ensemble

The final prediction for submission_improved combines both approaches via probability averaging:

```
final_preds = np.argmax(
    0.20 * rf_probs + 0.35 * lgb_probs + 0.45 * cnn_probs,
    axis=1
)
```

RF and LGB capture temporal dependencies through engineered statistics (autocorrelation, FFT, segment features, cross-axis correlations). The CNN captures learned temporal patterns directly from the raw 300-step sequence. Their ensemble benefits from diversity: when a tabular model misclassifies a borderline activity because its FFT features are ambiguous for a particular user, the CNN's direct temporal pattern recognition may correctly identify it, and vice versa.

The dominant confusion pairs are consistent across all methods: L2→L1 (48 – 58 of 72 class-2 validation instances are predicted as class 1) and L5→L1 (30 – 55 of 105 class-5 validation instances are predicted as class 1). Neither the tabular models nor the CNN fully solve the class-2 confusion, which is the most persistent problem in the project. The GroupKFold analysis showed that reducing CNN weight from 0.45 to 0.25 (submission_private_low_cnn_ensemble) improved cross-user generalization (public 0.7800) because the tabular models' hand-crafted features are more robust to user-level distribution shift than the CNN's learned patterns.

LightGBM 5-Fold OOF CV F1 (R1 = first try; R2 = re-run)

Method	Run	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	OOF mean

LGB + Residual CNN + augmented CNN ensemble	R1	0.7399	0.7100	0.7505	0.7410	0.7472	0.7380
LGB + Residual CNN + augmented CNN ensemble	R2	0.7300	0.7275	0.7101	0.7469	0.7307	0.7290
LightGBM 5-fold + single CNN	R1	0.7399	0.7100	0.7505	0.7410	0.7472	0.7380
LightGBM 5-fold + single CNN	R2	0.7300	0.7275	0.7101	0.7469	0.7307	0.7290
LGB + XGB + multi-seed CNN extended features (LGB)	R1	0.7548	0.7228	0.7551	0.7505	0.7737	0.7521
LGB + XGB + multi-seed CNN extended features (LGB)	R2	0.7537	0.7488	0.7171	0.7623	0.7636	0.7487
LGB + XGB + multi-seed CNN extended features (XGB)	R1	0.7425	0.7125	0.7388	0.7454	0.7574	0.7399
LGB + XGB + multi-seed CNN extended features (XGB)	R2	0.7559	0.7649	0.7108	0.7502	0.7402	0.7446
RF + LGB 5-fold + multi-seed CNN	R1	0.7336	0.7122	0.7416	0.7275	0.7432	0.7322
RF + LGB 5-fold + multi-seed CNN	R2	0.7335	0.7306	0.6984	0.7513	0.7366	0.7299
Person-independent LightGBM normalized features	R1	0.7380	0.7147	0.7387	0.7387	0.7492	0.7366
Person-independent LightGBM normalized features	R2	0.7515	0.7238	0.7020	0.7342	0.7443	0.7313

Section 4: Ablation Study

The ablation study tracks the contribution of individual design choices by examining score changes as components were added or modified across submissions. All public scores are from Kaggle's public leaderboard (macro-F1). OOF scores are from cross-validation on training data.

The project evolved through three distinct phases.

Phase 1 — Early exploration (submission-2 through submission_SMA): Starting from the RF-only baseline (0.7209), the first hybrid (submission-3, 0.7537) demonstrated the value of adding a CNN-LSTM. FFT features added on top of that (submission-5, 0.7227) unexpectedly dropped performance, and the SMA + peak-frequency variant (submission_SMA, 0.7475) only partially recovered. These early results revealed that adding frequency features to a simple CNN-LSTM with no cross-validation was inconsistent, as the single-seed training introduced high run-to-run variance.

Phase 2 — Major architecture upgrade (submission_improved through submission_v6): submission_improved replaced the CNN-LSTM with a Residual 1D-CNN, added LightGBM alongside Random Forest, and introduced a substantially richer feature set, reaching the overall best public score of 0.7881 — a gain of +0.0344 over submission-3. Subsequent versions (v2 – v6) explored cross-validation, multi-seed averaging, XGBoost stacking, and CNN weight adjustments, none of which surpassed submission_improved on the public leaderboard.

Phase 3 — Private generalization experiments: After observing the private leaderboard drop (ranking ~44th when v2 and v6 were selected as final submissions), targeted experiments addressed user-distribution shift: GroupKFold by user, reduced CNN weight, temperature scaling, and user-median normalization. These scored 0.7557 – 0.7800 on the public leaderboard. submission_private_low_cnn_ensemble (0.7800) became the best among these, confirming that reducing CNN weight and applying GroupKFold training improves cross-user generalization.

4.1 Feature Engineering Ablation

Configuration	Public Score	Delta vs. previous row
Baseline: RF-only, 30 features (global stats + rolling mean + magnitude)	0.7209	—
RF + CNN-LSTM, same global-stat features (submission-3)	0.7537	+0.0328
RF + CNN-LSTM + FFT energy, max, mean per axis (submission-5)	0.7227	−0.0310
RF + CNN-LSTM + SMA + peak frequency + spectral entropy, replacing full FFT (submission_SMA)	0.7475	+0.0248
Full rebuild: RF + LGB + Residual 1D-CNN, complete rich feature set: autocorr lags, IQR, skew, kurtosis, ZCR, p10/p90, segment stats, top-3 FFT, cross-axis corr, tilt (submission_improved)	0.7881	+0.0406

Note: The last row reflects a complete pipeline rebuild (new architecture, new tabular model, new feature set), not an incremental addition to submission_SMA. The +0.0406 delta reflects the combined effect of all three changes.

Observation: Simply adding FFT energy features to the CNN-LSTM baseline degraded performance (−0.0310). There are several reasons for this. First, FFT energy/max/mean computed over the full 300-row window are coarse summaries that are at least partially redundant with what the CNN-LSTM already captures from the raw sequence — the CNN sees the raw signal and implicitly learns spectral characteristics, so adding redundant hand-crafted spectral features adds noise without adding independent discriminative signal. Second, total spectral energy is dominated by low-frequency components and is sensitive to baseline drift between users; it varies more between users doing the same activity than between activities within the same user, making it a noisy feature for a classifier trained without user-level normalization. Third, these three new features (energy, max, mean) are computed only on the tabular RF side, not integrated into the CNN. They expand the RF's feature space slightly while diluting the relative importance of the already-informative global stats, potentially disrupting the RF's existing decision boundaries without providing enough new signal to compensate.

Switching to SMA + peak frequency + spectral entropy (submission_SMA) recovered significantly (+0.0248 vs. submission-5). Peak frequency is semantically stable: it identifies the dominant repetition rate of the activity, which is consistent across users performing the same activity (e.g., walking always produces a dominant step frequency regardless of the user's arm length or wearing angle). Spectral entropy captures whether

the spectrum is concentrated (periodic activity) or diffuse (random motion), which is also user-invariant. These features address a different dimension of the signal than global statistics, adding genuine complementary information rather than noisy redundancy.

The most important jump came from the complete rebuild in submission_improved (+0.0406). The gain comes from three simultaneous sources that cannot be individually isolated from this table alone: the feature enrichment (autocorrelation at multiple lags captures rhythm periodicity; IQR, skew, and kurtosis capture distributional shape beyond mean and variance; temporal segment stats capture within-recording non-stationarity; cross-axis correlations capture motion geometry), the addition of LightGBM (which uses gradient boosting and handles feature interactions differently from RF's bagging approach, reducing correlated errors), and the switch to Residual 1D-CNN (which processes the full 300-step sequence end-to-end rather than 10 coarse windows). All three changes compound.

4.2 Model Architecture Ablation

Architecture	Public Score	Delta vs. previous row
RF only, global-stat features (submission-2)	0.7209	—
RF + CNN-LSTM, global-stat features (submission-3)	0.7537	+0.0328
LGB 5-fold + single Residual 1D-CNN, rich features (v3)	0.7770	+0.0233
RF + LGB + Residual 1D-CNN, full rich features (submission_improved)	0.7881	+0.0111
LGB + XGB (minor) + 3-seed CNN (v4)	0.7547	−0.0223 vs. v3
Bigger CNN + label smoothing + spatial dropout (v10, seed42)	0.7320	−0.0227 vs. v4

Observation: The +0.0233 gain from submission-3 to v3 reflects two simultaneous changes that cannot be cleanly separated: the switch from CNN-LSTM to Residual 1D-CNN, and the addition of LightGBM alongside RF. The architectural change likely accounts for the majority of the gain. The CNN-LSTM segmented the 300-row sequence into 10 coarse 30-second windows before modeling temporal dependencies, meaning any pattern shorter than ~30 seconds was averaged away before the LSTM saw it. The Residual 1D-CNN processes the full flat 300-timestep sequence with convolutional kernels of width 5 – 7, giving it access to all temporal scales simultaneously. Residual skip connections allow gradients to flow directly to early layers, preventing the vanishing gradient problem that

limits plain CNNs on long sequences. Multi-scale global pooling (average + max concatenated) captures both the typical activity signature across the full recording and the most distinctive moment — the max pool is particularly useful for activities with brief but characteristic motion bursts that would be diluted by averaging.

Adding RF back alongside LGB in submission_improved yielded a further +0.0111 public score gain. It should be noted that this transition also involved moving from v3's feature set to the full rich 69-feature set used in submission_improved, so the +0.0111 delta cannot be attributed to the RF addition alone — it reflects both the extra model diversity and any feature differences between the two versions. Within that caveat, the RF contribution is meaningful because RF uses bagging on random feature subsets while LGB uses sequential boosting on the full feature set, producing prediction errors that are partially decorrelated: on borderline instances where LGB's boosted trees have overfit to a specific feature interaction, RF's averaging over many independent trees may produce a more conservative and correct probability. The optimal ensemble weights (RF=0.20, LGB=0.35, CNN=0.45) reflect that RF contributes the least unique signal of the three components, but its 0.20 weight still meaningfully diversifies the tabular half of the ensemble relative to LGB alone.

Adding XGBoost in v4 did not improve performance (0.7547 vs. 0.7770 in v3, a drop of −0.0223). The optimal ensemble blend found by the OOF weight search was LGB=0.40, XGB=0.10, CNN=0.50 — XGB received only a 0.10 weight, indicating it contributed almost nothing beyond what LGB already captured. The reason is that XGB and LGB are both gradient-boosted decision trees trained on the same 133-feature set, making their predictions highly correlated: the XGB-only OOF macro-F1 was 0.7399 versus LGB-only OOF of 0.7521, confirming XGB was simply the weaker model rather than a complementary one. When LGB misclassifies a borderline instance because certain features are ambiguous for that user, XGB trained on the same features will make the same error. The ensemble therefore averaged a stronger model with a weaker correlated one rather than combining complementary strengths. The OOF ensemble proxy score of 0.7494 looked promising, but this was optimistic because the CNN component — which received the highest weight of 0.50 — was not evaluated in a proper cross-validated OOF structure, inflating the apparent benefit of high CNN weighting. The public score of 0.7547 revealed the true cost.

The bigger CNN with label smoothing and spatial dropout (v10) degraded performance by −0.0227 vs. v4. Label smoothing redistributes probability mass away from the true class (e.g., 0.9 for the correct label instead of 1.0), which is harmful here because the minority classes (2 and 5) are already underrepresented and require sharp decision boundaries —

softening the targets makes the model treat a correct minority-class prediction as nearly as good as a wrong one, degrading recall on exactly the classes that dominate macro-F1. The per-class validation F1 data confirms this: the best LGB component achieved class-2 F1 of only 0.1798 and class-5 F1 of 0.6395 even without label smoothing, and both are the dominant confusion targets (L2→L1 accounts for 48 – 58 of 72 class-2 validation instances across all methods; L5→L1 accounts for 30 – 55 of 105 class-5 instances). Softening the training signal on these already-difficult classes further reduces their recall without any compensating benefit. Spatial dropout drops entire convolutional feature map channels during training. For majority classes with thousands of examples this acts as useful regularization, but for class 4 (142 samples) and class 2 (358 samples) losing entire feature channels in each forward pass prevents the network from reliably learning the sparse signatures these classes produce, effectively amplifying the sample scarcity problem rather than mitigating it.

4.3 Cross-Validation Strategy Ablation

CV Strategy	OOF F1-macro	Public Score	Notes
No CV (single train/val split, ~90/10)	N/A	0.7537 (submission-3)	Single seed, fixed 20 epochs
StratifiedKFold 5-fold, LGB-only OOF (v3)	0.7380	0.7770	LGB-only OOF; +0.0233 vs. no CV
StratifiedKFold 5-fold, ensemble OOF (v4)	0.7494	0.7547	Best-blend OOF (LGB=0.40, XGB=0.10, CNN=0.50); LGB-only OOF was 0.7521
GroupKFold 5-fold by user (private LGB only)	0.7123	0.7717	More honest, stable cross-user
GroupKFold 5-fold by user (private GKF ensemble)	0.7167	0.7744	RF+LGB+CNN ensemble under GroupKFold

Observation: The OOF column is not directly comparable across v3 and v4 — v3 reports LGB-only OOF (0.7380) while v4 reports the best ensemble blend OOF (0.7494, at LGB=0.40, XGB=0.10, CNN=0.50). v4's LGB-only OOF was actually 0.7521, higher than v3's 0.7380, because v4 used a richer 133-feature set. XGB-only OOF was 0.7399. Despite the higher OOF, v4's public score (0.7547) was substantially lower than v3 (0.7770), a drop of −0.0223.

There are several reasons why a higher ensemble OOF did not translate to a better public score. First, the CNN component (weight=0.50 in the optimal blend) was trained without cross-validation on the OOF search — the CNN predictions used in the weight sweep were in-sample or from a single fixed split, making the OOF proxy optimistic for any blend that up-weights the CNN. Second, the three CNN seeds (42, 7, 123) were averaged but not evaluated in a proper held-out fold structure, so the 0.7494 OOF proxy inflated the apparent benefit of high CNN weight. Third, adding XGBoost provided minimal genuine diversity over LGB (both are gradient-boosted trees on the same 133 features; XGB OOF 0.7399 vs. LGB OOF 0.7521 shows XGB was simply the weaker model), meaning the ensemble mostly averaged a stronger model with a weaker correlated one rather than combining complementary strengths.

StratifiedKFold still substantially outperforms no-CV for the LGB component (0.7521 vs. the implied single-split performance of submission-3's RF). The fundamental limitation of StratifiedKFold — shuffling files across users so the model sees some recordings from every user during training — applies equally to v3 and v4, which is why both produce optimistically high OOF scores relative to GroupKFold's 0.7123.

4.4 Ensemble Weight Ablation

Configuration	CNN Weight	OOF / Public Score	Notes
RF+LGB tabular blend only (GKF, best sweep)	0.00	OOF 0.7133	w_rf=0.35, w_lgb=0.65; no public submission
LGB only, GroupKFold	0.00	OOF 0.7123 / public 0.7717	Tabular-only public baseline
Private low-CNN ensemble	0.25	public 0.7800	Best private-aware submission; weights RF=0.25, LGB=0.50, CNN=0.25
submission_improved	0.45	public 0.7881	Best public score overall
submission_v6	0.55	public 0.7664	−0.0217 vs. CNN=0.45

Observation: The optimal CNN weight was 0.45 for the public leaderboard, but reducing it to 0.25 (used in submission_private_low_cnn_ensemble) was better for private

generalization. The GroupKFold OOF weight sweep separately found an optimal 3-way split at RF=0.20, LGB=0.50, CNN=0.30 (OOF 0.7167), a closely related configuration. As CNN weight increases beyond 0.45, user-specific temporal patterns that the CNN memorized during training dominate the ensemble, degrading cross-user robustness. The tabular models (RF, LGB) rely on hand-crafted features with built-in invariances (autocorrelation, spectral entropy, cross-axis correlations), making them more robust to user-level distribution shift. The optimal ensemble weight is therefore a tradeoff between exploiting the CNN's temporal pattern-matching strength and limiting its user-overfitting tendency.

4.5 Training Stability Ablation (Multi-Seed and Augmentation)

All Phase 2 submissions were re-run under identical code to measure run-to-run variance. submission_seed2 (seed=2 variant of the submission_improved architecture) and submission_v10_seed0 (seed=0 variant of v10_seed42) provide additional seed comparison points. The full rerun record is:

Submission	Original Score	Rerun Score	Delta	Key Factor
submission_improved	0.7881	0.7614 (submission_improved-2)	−0.0267	Largest drop; single seed, no CV averaging
submission_v2	0.7676	0.7526 (submission_v2-2)	−0.0150	Augmentation increased variance
submission_v3	0.7770	0.7654 (submission_v3-2)	−0.0116	Most stable Phase 2 method
submission_v4	0.7547	0.7678 (submission_v4-2)	+0.0131	Variance can go either direction
submission_v5	0.7484	0.7342 (submission_v5-2)	−0.0142	Multi-seed CNN did not reduce variance
submission_v6	0.7664	0.7475 (submission_v6-2)	−0.0189	High CNN weight amplified instability

submission_v10_seed 42	0.7320	—	—	Seed search only; seed=0 variant below
submission_v10_seed 0	0.7469	0.7521 (submission_seed0-2)	+0.005 2	Small positive variance
submission_seed2	0.7626	—	—	Seed=2 variant of submission_improv ed architecture

Observation: Score variability across reruns is not a simple downward trend — v4 improved from 0.7547 to 0.7678 on rerun, and v10_seed0 improved slightly from 0.7469 to 0.7521, showing that stochastic variation can go either direction. The sources of non-determinism are: GPU-level cuDNN nondeterminism (not eliminated by Python/TF seeds without TF_DETERMINISTIC_OPS=1), mini-batch ordering (shuffle=True in Keras fit), and Keras's random 10% validation split (no fixed seed), which changes which weights get selected by EarlyStopping.

The most striking rerun result is submission_improved, which dropped −0.0267 on rerun — the largest variance of any submission. Despite having the highest single-run public score (0.7881), this makes it the least reliable indicator of true model quality. The three seed variants of the submission_improved architecture — seed=0 (original, 0.7881), seed=2 (submission_seed2, 0.7626), and the rerun (submission_improved-2, 0.7614) — span a range of 0.0267, indicating the headline score was likely a fortunate seed draw. In contrast, v3 dropped only −0.0116, confirming that fewer CNN degrees of freedom (lower CNN weight, single seed, no augmentation) produces more reproducible results.

The augmentation applied in v2 (time-scaling of the 1-second summary rows) was conceptually flawed: stretching or compressing rows of pre-aggregated mean/std statistics does not correspond to any realistic physical variation in wrist accelerometer data, so it introduced distributional noise that made the CNN less discriminative while simultaneously increasing run-to-run variance (−0.0150 drop, vs. −0.0116 for un-augmented v3).

Based on the full rerun data, the observed score variation across independent runs ranges from −0.0267 to +0.0131, meaning models that differ by less than ~0.010 – 0.015 on a single evaluation cannot be reliably ranked. The private GKF ensemble (submission_private_gkf_ensemble, 0.7744) was trained with 5 GroupKFold folds, each using a single CNN seed, with fold predictions averaged at test time — this reduces fold-

level variance but does not sweep multiple seeds per fold. It was not re-run separately, but its GroupKFold structure makes it inherently more stable than single-split methods.

4.6 Public-Private Leaderboard Gap Analysis

The private leaderboard revealed an important generalization failure. When v2 and v6 were selected as final private submissions, the private ranking dropped to approximately 44th place — below competitors whose public scores were as low as 0.7448, meaning models with substantially lower public scores generalized better to the private test set. By contrast, when submission_improved and submission_v3 were selected as final submissions in the prior week, the private ranking was 23rd. This reversal illustrates that public leaderboard score is not a reliable proxy for private generalization when the test distribution includes entirely new users.

Their rerun scores (v2: 0.7526, v6: 0.7475) indicate both were near the lower end of their true performance distributions, compounding the generalization penalty.

Submission	Public Score	Rerun Score	Key Risk Factor
submission_improved	0.7881	0.7614 (−0.0267)	Largest seed variance; CNN weight 0.45
submission_v3	0.7770	0.7654 (−0.0116)	Most stable; lower CNN weight, no augmentation
submission_v2	0.7676	0.7526 (−0.0150)	Unrealistic augmentation increased variance
submission_v6	0.7664	0.7475 (−0.0189)	CNN weight 0.55 amplified overfitting

The root cause of the private drop is user distribution shift. The 60 training users and the public/private test users are completely separate people. A CNN trained with a high weight in the ensemble will learn user-specific temporal patterns — the exact cadence, wrist tilt trajectory, and transition timing that characterize individual people — in addition to activity-class patterns. These user-specific patterns do not transfer to new users and actively hurt predictions when the test person's motion style differs from all 60 training users. v2's unrealistic augmentation (stretching and compressing rows of pre-aggregated mean/std statistics) distorted the CNN's learned feature distribution in a direction that does not correspond to any real physical variation, adding noise that degraded generalization without improving robustness. v6's CNN weight of 0.55 gave the most user-overfitting

component the most influence over the final prediction. A better selection criterion would have been: prefer models with lower CNN weight, trained under GroupKFold by user, confirmed stable across at least two independent reruns — all three conditions that submission_private_low_cnn_ensemble satisfied.

Summary of All Submissions

Submission	Key Design	Public Score	Key Observation
submission-2	RF-only, 30 global-stat features	0.7209	Naive baseline; statistics alone insufficient
submission-3	RF + CNN-LSTM, 26-col global stats	0.7537	+0.0328 from adding temporal sequence model
submission-5	RF + CNN-LSTM + FFT energy/max/mean	0.7227	FFT energy features hurt RF; −0.0310 regression
submission_SMA	RF + CNN-LSTM + SMA + peak freq	0.7475	Peak freq more stable than full FFT energy
submission_improved	RF + LGB + Res-CNN, full rich features	0.7881	Best public score; rerun dropped to 0.7614 (−0.0267)
submission_v3	LGB 5-fold + single Res-CNN	0.7770	Most stable Phase 2 method; rerun −0.0116

submission_v2	CNN aug (time-scale) + 2 seeds	0.7676	Augmentation flawed; rerun dropped to 0.7526
submission_v4	LGB + XGB + 3-seed CNN	0.7547	XGB correlated with LGB; rerun improved to 0.7678
submission_v5	RF + LGB 5-fold + multi-seed CNN (RF=0.15, LGB=0.30, CNN=0.55)	0.7484	Rerun dropped to 0.7342; CNN weight too high
submission_v6	CNN weight 0.45 → 0.55	0.7664	More CNN weight amplified user overfitting; rerun 0.7475
submission_v10_seed42	Bigger CNN + label smooth + spatial dropout	0.7320	Over-regularized; label smoothing hurt minority classes
submission_v10_seed0	Same architecture, seed=0	0.7469	+0.0149 vs. seed=42; seed variance alone spans 0.0149
submission_private_lgb_groupkfold	GroupKFold LGB only, no CNN	0.7717	Tabular-only competitive; GKF OOF = 0.7123
submission_private_low_cnn_ensemble	GroupKFold + RF/LGB/CNN=0.25/0.50/0.25	0.7800	Best private-aware submission; lower CNN weight helps

submission_private_tabular_blend	OOF weight sweep for RF and LGB probabilities	0.7653	RF/LGB blend; underperformed pure LGB (0.7717), showing RF addition hurts.
submission_private_lgb_xgb_stack	GroupKFold LGB + XGB blend	0.7714	Best blend OOF 0.7140 (w_lgb=0.40, w_xgb=0.60)
submission_private_gkf_ensemble	GroupKFold 5 folds, single CNN seed per fold, averaged	0.7744	GroupKFold structure reduces fold variance; GKF OOF CNN = 0.6772; GKF ensemble OOF = 0.7167
submission_private_calibrated	Temperature scaling on RF+LGB tabular blend	0.7655	Temperatures near 1.0; OOF F1 unchanged (0.7133→0.7129)
submission_private_user_median_lgb	User-median normalization + LGB	0.7557	Train/test identity mismatch; OOF 0.7206
submission_seed2	submission_improved architecture, seed=2	0.7626	−0.0255 vs. seed=0 original; confirms wide seed variance
submission_improved-2	submission_improved rerun (seed=0)	0.7614	−0.0267 vs. original; largest rerun drop

submission_v2-2	submission_v2 rerun	0.7526	−0.0150; augmentation increased variance confirmed
submission_v3-2	submission_v3 rerun	0.7654	−0.0116; most stable Phase 2 method
submission_v4-2	submission_v4 rerun	0.7678	+0.0131; stochastic variance can go either direction
submission_v5-2	submission_v5 rerun	0.7342	−0.0142; multi-seed CNN did not reduce variance
submission_v6-2	submission_v6 rerun	0.7475	−0.0189; high CNN weight amplified instability
submission_seed0-2	submission_v10_seed0 rerun	0.7521	+0.0052 vs. original